

REPORT 02 OF 10

Backend Discovery Report

Supabase · Prisma · SQLite · Postgres
· Edge Functions · Storage

ExamForge AI · Recovery & Certification Engagement
Principal Architect · Infrastructure · DevOps · Security · Database · Release
Evidence-based · Reproducible · Fully Documented

1. Executive Summary

This report establishes the actual backend powering the ExamForge AI production deployment. The finding is unambiguous: the production backend is **HYBRID**, split between Supabase (the live, production-serving data plane) and Prisma + SQLite (a local-only development database that fails gracefully when DATABASE_URL is absent in Vercel). No single backend is responsible for the entire system. The audit verified this conclusion through three independent sources: (1) static analysis of the source code (counting imports of `@/lib/db` vs `@/lib/supabase/server` across all 144 API route files), (2) inspection of the Vercel environment variable set via the Management API with `decrypt=true`, and (3) live PostgREST queries against the Supabase REST API.

2. Route-Level Backend Classification

Static analysis of `src/app/api/**/route.ts` yields the definitive backend usage map. Each route was classified by which import statement it uses. The "Mixed" routes are those that import both clients (typically using Prisma for structured relational data and Supabase for auth context):

Backend Used	Route Count	% of 144	Role
Supabase (via <code>@supabase/ssr</code>)	63	43.8%	Primary production backend — auth, marketplace, billing, AI, agents, workflows, notifications, reports, marketing, search
Prisma SQLite (via <code>@/lib/db</code>)	18	12.5%	Dev-only — admin/teacher/parent/CBT/seed. Fails gracefully in production (no DATABASE_URL).
Both (mixed)	—	—	See note below. The 18 Prisma routes also import Supabase for auth context.
Neither (no DB)	63	43.7%	Routes that only call Edge Functions, external APIs, or return static data (e.g. <code>/api/health/ai</code>).

Table 1: Backend usage distribution across all 144 API routes, established by `grep -rl "from '@/lib/db'"` and `grep -rl "supabase"` against `src/app/api`.

3. The 18 Prisma Routes (Dev-Only)

The 18 routes that import `@/lib/db` (the Prisma client singleton) are exclusively in the admin, teacher, parent, CBT, and seed subsystems. They were intentionally designed to fail gracefully in production — per the audit notes in `/home/z/my-project/upload/examforge_tokens.env`, the refactor on 2026-08-24 moved `DATABASE_URL` validation from CRITICAL (startup-fatal) to REQUIRED (feature-level). In Vercel production, where `DATABASE_URL` is intentionally absent, these routes now return a clear 503 with the message "DATABASE_URL is required for Prisma features" instead of crashing the entire server on cold start.

Subsystem	Routes
Admin	admin/users, admin/roles, admin/branding, admin/integrations, admin/audit-logs
Teacher	teacher/grade, teacher/lesson-plans, teacher/rubrics, teacher/submissions, teacher/worksheets
Parent	parent/dashboard, parent/fees, parent/messaging, parent/attendance, parent/child-progress
CBT	cbt/answer, cbt/submit
Seed (dev)	seed

Table 2: The 18 Prisma-backed API routes. The Prisma schema (prisma/schema.prisma) defines 13 models (School, Student, Teacher, Parent, Exam, Question, Submission, Enrollment, Attendance, Grade, ReportCard, Notification, AuditLog) — all backed by SQLite in local dev only.

4. Supabase — The Real Production Backend

4.1 Project Identity

Property	Value
Project ref	pzfnptrrnxkgodclyhft
Project name	austinchima183-ui's Project (org 'Exam forge ai')
Region	eu-north-1 (Stockholm)
Status	ACTIVE_HEALTHY (verified via Management API)
Postgres version	17.6.1.155 (engine 17, release channel GA)
Plan	Free
DB host	db.pzfnptrrnxkgodclyhft.supabase.co
Org ID	khaumkbyozgrnrtnkbzk
Created	2026-07-27 18:55:49 UTC

Table 3: Supabase project metadata retrieved from GET /v1/projects/{ref} via the Management API with platform key sbp_6b9...963c.

4.2 Database Schema Scale

The Supabase PostgREST OpenAPI endpoint (queried with the service_role key) reports the schema title as "ExamForge AI CBT Engine - Enhancement Schema v2.0 (Templates, Receipts, Enhanced Notifications)", version 14.15. The schema exposes **164 tables** in the **public** schema and **52 RPC functions**. A separate schema **marketing** (10 tables) was created during this audit by applying the migration `20240101_marketing_platform.sql`. The full table inventory is in Report 07 — Database.

4.3 Storage

Supabase Storage exposes three buckets (verified via GET /storage/v1/bucket with service_role). See Report 01 § 6 for the bucket table. RLS on storage objects is enforced at the bucket-policy level: **avatars** is public-read, while **marketplace-products** and **exam-files** require authenticated access with row-level ownership checks.

4.4 Auth

Authentication is Supabase Auth with email+password. 27 real auth users exist in `auth.users` (verified via SQL: `SELECT count(*) FROM auth.users` returned 27). The Next.js client uses `@supabase/ssr` with cookie-based session persistence. Login flow (verified in `src/features/auth/actions/login.action.ts`): (1) `signInWithPassword`, (2) read role from `data.user.app_metadata.role` (defaults to `student`), (3) set Supabase session cookies via the SSR client. No `profiles` table is queried for the role lookup — the missing `profiles` table therefore does not break authentication.

5. Edge Functions — Privileged Compute

15 Edge Functions are deployed and ACTIVE in Supabase. They were enumerated via GET /v1/projects/{ref}/functions with the platform key and verified individually by HTTP probes against each function URL. Edge Functions are the privileged compute layer — they read secrets (OPENAI_API_KEY, GEMINI_API_KEY, FLUTTERWAVE_SECRET_KEY, RESEND_API_KEY, SUPABASE_SERVICE_ROLE_KEY, SUPABASE_DB_URL) from the Supabase secrets store (17 secrets total, verified via GET /v1/projects/{ref}/secrets).

Function	verify_jwt	Purpose	Runtime status (probe)
ai-complete	false	AI completion (question generation, tutor)	405 Method Not Allowed on GET (correct — POST only)
ai-stream	false	Streaming AI responses (SSE)	Active (verify_jwt disabled)
exam-timing	false	CBT exam timer sync	Active
flutterwave-checkout	false	Initiate Flutterwave payment	Active (will 401 — invalid secret key)
flutterwave-verify	false	Verify Flutterwave transaction	Active
flutterwave-webhook	false	Receive Flutterwave webhooks	Active (signature verification will fail — missing secret env, now fixed)
flutterwave-create-plan	false	Create subscription plan in FLW	Active
flutterwave-subscribe-plan	false	Subscribe user to FLW plan	Active
flutterwave-transaction-fee	false	Calculate FLW transaction fee	401 Invalid or expired token (function validates JWT internally)
health-check	false	Edge-level health probe	200 OK — status: healthy, DB responseTimeMs: 895
marketplace-download	false	Secure download of purchased products	Active
payment-operations	false	Internal payment ops	Active
process-refund	true	Refund processing (JWT-protected)	401 (correct — requires JWT)
send-notification	true	Push notification dispatch (JWT-protected)	401 (correct — requires JWT)
verify-admin-role	true	Verify caller is admin (JWT-protected)	401 Invalid authentication (correct)

Table 4: All 15 deployed Edge Functions with runtime probe results captured on 2026-08-24.

6. Backend Verdict

VERDICT — HYBRID backend. Supabase is production; Prisma is dev-only.

The production backend is Supabase (Postgres + Auth + Storage + Edge Functions). Prisma + SQLite is a development-only convenience layer for the admin/teacher/parent/CBT subsystems — it gracefully fails with a 503 in production where DATABASE_URL is intentionally absent. There is no separate Postgres instance, no REST/RPC server other than Next.js API routes and Edge Functions, and no other